

THE $n = 10$ CASE OF A q -FIBONACCI PRODUCT CONJECTURE FOR KAZHDAN–LUSZTIG R -POLYNOMIALS

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ABSTRACT. Chen, Fan, Guo, and Zhong conjectured that, for $e \leq u \leq v \leq v_n = 34 \cdots n12$, the modified Kazhdan–Lusztig R -polynomial is a power of q times a product of q -Fibonacci polynomials. They reported a verification through $n = 9$. We give an exact, independently checkable finite certification of the first subsequent case, $n = 10$. The interval $[e, v_{10}]$ has 12,866 elements and contains 6,229,297 comparable ordered pairs; every pair satisfies the product formula. The certificate uses integer arithmetic, regenerates the complete universe, decides membership in the full Fibonacci multiplicative monoid, and cross-checks two recurrences, one for $\tilde{R}$ and one for the ordinary R -polynomials. We also prove support-factorization and capped-ladder reductions that isolate the genuinely new connected cases. This is a certified finite result, not a proof of the conjecture for arbitrary n .

1. INTRODUCTION

Kazhdan and Lusztig introduced the polynomials now bearing their names in the study of Hecke algebras and Coxeter groups [4]. We use standard facts about reduced words and strong Bruhat order as in Björner and Brenti [1]. Dyer’s modified polynomials $\tilde{R}_{u,v}$ have nonnegative coefficients and obey a particularly simple recurrence [3].

Put

$$v_n = [3, 4, \dots, n, 1, 2] \in S_n.$$

Chen, Fan, Guo, and Zhong conjectured that every $\tilde{R}_{u,v}$ with $e \leq u \leq v \leq v_n$ is a power of q times a product of q -Fibonacci polynomials. Their manuscript reports verification for $n \leq 9$ and proves the case in which the upper endpoint equals v_n [2]; the identity at the lower endpoint e originates in Pagliacci’s formula [5].

The purpose of this note is narrow and exact: certify the complete $n = 10$ instance. Searches recorded with the accompanying files found no journal

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version of the 2013 manuscript and no public resolution of this exact conjecture as of August 29, 2026. This is a bounded literature statement, not a claim that every database is complete.

Our principal result is the following.

Theorem 1.1 (Certified finite result). *For every $u, v \in S_{10}$ satisfying*

$$e \leq u \leq v \leq [3, 4, 5, 6, 7, 8, 9, 10, 1, 2],$$

there exist $k \geq 0$ and integers $h_1, \dots, h_k \geq 2$ such that

$$\tilde{R}_{u,v}(q) = q^{\ell(v)-\ell(u)} \prod_{i=1}^k F_{h_i}(q^{-2}).$$

The product is empty when $u = v$. The assertion is certified for all 6,229,297 comparable ordered pairs in the interval.

The word “certified” means that a small serialized input and a standalone fail-closed verifier reconstruct the finite objects and exact recurrences. No floating-point calculation, modular proxy, random sample, or imported table of discovery polynomials enters the proof.

2. DEFINITIONS AND THE FORCED NORMALIZATION

Permutations are written in one-line notation and multiplied on the right; ws_i swaps positions i and $i + 1$. The length $\ell(w)$ is the inversion number. We use the rank-matrix criterion

$$(2.1) \quad u \leq v \iff \#\{i \leq p : u(i) \leq r\} \geq \#\{i \leq p : v(i) \leq r\} \quad (1 \leq p, r \leq n).$$

and the subword property in their standard type- A forms [1, Chapter 2]. The fixed reduced word

$$(2.2) \quad \Omega_n = s_2 s_1 s_3 s_2 \cdots s_{n-1} s_{n-2}$$

represents v_n and has length $2n - 4$.

For incomparable endpoints set $\tilde{R}_{u,v} = 0$, and set $\tilde{R}_{u,u} = 1$. If $u < v$ and $s \in D_R(v)$, then

$$(2.3) \quad \tilde{R}_{u,v}(q) = \begin{cases} \tilde{R}_{us,vs}(q), & s \in D_R(u), \\ \tilde{R}_{us,vs}(q) + q\tilde{R}_{u,vs}(q), & s \notin D_R(u). \end{cases}$$

in the normalization used by Chen–Fan–Guo–Zhong [2, Theorem 1.6]; the modified-polynomial provenance is Dyer’s [3]. Define

$$(2.4) \quad F_0(x) = F_1(x) = 1, \quad F_h(x) = F_{h-1}(x) + xF_{h-2}(x) \quad (h \geq 2).$$

as in the source conjecture [2, p. 3]. The factors F_0 and F_1 are units and are deleted from canonical factor lists.

Let $d = \ell(v) - \ell(u)$. Induction in (2.3) shows that $\tilde{R}_{u,v}$ is monic of degree d and is supported in degrees congruent to d modulo two. Hence there is a unique $B_{u,v}(x) \in \mathbb{Z}[x]$ such that

$$(2.5) \quad \tilde{R}_{u,v}(q) = q^d B_{u,v}(q^{-2}).$$

Because each F_h has constant term one, the exponent in the conjectural formula is forced to be d ; it is not the q -adic valuation. After (2.5), the conjecture becomes the exact membership problem

$$(2.6) \quad B_{u,v}(x) \in \langle F_2(x), F_3(x), \dots \rangle_{\text{mult}}.$$

This formulation is important: an allowed F_h can be reducible. For example,

$$F_5(x) = 1 + 4x + 3x^2 = (1 + x)(1 + 3x).$$

Thus requiring every irreducible factor to equal an F_h would reject a known valid case.

Dividing (2.3) by q^d gives the recurrence used by the first evaluator:

$$(2.7) \quad B_{u,v} = \begin{cases} B_{us,vs}, & s \in D_R(u), \\ B_{u,vs} + xB_{us,vs}, & s \notin D_R(u), \end{cases}$$

where a term with incomparable endpoints equals zero.

3. STRUCTURAL REDUCTIONS

These reductions are not needed to trust the exhaustive certificate, but they separate products forced by independent Coxeter supports from the new connected cases.

3.1. Support products. Let $\text{Supp}(w)$ be the set of simple generators occurring in a reduced expression of w . If the maximal consecutive components of $\text{Supp}(v)$ are $I_1, \dots, I_r$, then the corresponding parabolic subgroups commute.

Proposition 3.1 (Support multiplicativity). *Suppose $u \leq v$ and write uniquely*

$$v = v_1 \cdots v_r, \quad u = u_1 \cdots u_r, \quad u_j, v_j \in W_{I_j}.$$

Then $u_j \leq v_j$ and

$$\tilde{R}_{u,v}(q) = \prod_{j=1}^r \tilde{R}_{u_j,v_j}(q), \quad B_{u,v}(x) = \prod_{j=1}^r B_{u_j,v_j}(x).$$

Proof. The subword property gives $\text{Supp}(u) \subseteq \text{Supp}(v)$. Bruhat order and length are coordinatewise in the direct product of the parabolics. Induct on $\ell(v)$ and choose a right descent in one component. All other components commute with that descent, so both branches of (2.3) extract the unchanged factors. The remaining bracket is exactly the recurrence in the chosen component. Length additivity gives the normalized identity. $\square$

If an upper component is boolean, meaning that a reduced word uses each simple generator at most once, (2.3) gives $\tilde{R}_{a,b} = q^{\ell(b) - \ell(a)}$ for every $a \leq b$. If a component is a shift of the full block v_m , the proved upper-endpoint formula of Chen–Fan–Guo–Zhong [2, Corollary 2.3] supplies one Fibonacci factor. Proposition 3.1 therefore proves the conjecture for every interval whose upper support components are boolean or shifted full blocks. In this class, the factor indices satisfy $\sum_i h_i \leq n - 2$.

3.2. The capped-ladder boundary at $n = 10$. For a connected support interval $I = [a, b]$, put

$$r = b - a + 1, \quad c_L = \mathbf{1}_{a>1}, \quad c_R = \mathbf{1}_{b<n-1}.$$

Projecting a subword of Ω_n onto I gives the local word

$$(3.1) \quad \Lambda_{a,b}^{(n)} = (s_a)^{c_L} \left(\prod_{j=a+1}^b s_j s_{j-1} \right) (s_b)^{c_R},$$

where an exponent records presence or absence. After shifting generator indices, this local interval occurs inside $[e, v_N]$ with

$$(3.2) \quad N = r + 1 + c_L + c_R.$$

Indeed, restricting Ω_N to the shifted support gives (3.1), and the subword property completes the embedding.

Consequently, once $n \leq 9$ is known, only three connected ambient families are new at $n = 10$:

	type	support	caps
(3.3)	A	$[1, 9]$	0
	B	$[1, 8]$ or $[2, 9]$	1
	C	$[2, 8]$	2.

The two type- B cases are exchanged by inverse followed by the type- A diagram involution. All disconnected-support cases factor into components of ambient rank at most nine. This reduction explains why the first new finite test is concentrated in a small capped-ladder boundary, even though the released verifier checks the entire interval.

4. EXACT FINITE CERTIFICATE

4.1. Complete universe and pair set. The primary verifier enumerates all of S_n lexicographically and filters by (2.1). Independently, it forms all products of subwords of (2.2). The two sets must agree exactly. It then applies (2.1) to every ordered pair in the set. Table 1 records the results; in particular, it reproduces the authors' reported $n \leq 9$ baseline before proceeding to $n = 10$.

4.2. Complete Fibonacci-monoid decision. For $n = 10$, (2.5) gives $\deg B_{u,v} \leq 8$. Since

$$\deg F_h = \lfloor h/2 \rfloor,$$

every nonunit factor in a possible representation has $2 \leq h \leq 17$. Conversely, recursively enumerating all nondecreasing multisets of these indices whose degree sum is at most eight enumerates every possible product. Exact multiplication produces 434 distinct coefficient vectors, including the unit. This is a complete finite monoid, not a greedy division or an irreducible-factor heuristic.

Upper endpoints are processed by increasing length. Formula (2.7) reconstructs each new exact coefficient vector from already verified lower ranks. A

TABLE 1. Complete exact enumeration.

n	$ [e, v_n] $	comparable ordered pairs	rejected pairs
2	1	1	0
3	4	9	0
4	14	69	0
5	46	493	0
6	146	3,385	0
7	454	22,657	0
8	1,394	149,021	0
9	4,246	967,941	0
10	12,866	6,229,297	0

vector absent from the finite monoid would stop the run and print an exact counterexample. No absence occurs.

4.3. A second recurrence. The verifier also computes the ordinary R -polynomials independently. In the non-descent branch it uses

$$(4.1) \quad R_{u,v}(Q) = QR_{us,vs}(Q) + (Q - 1)R_{u,vs}(Q).$$

from the ordinary recurrence [4, Section 2]. For every gap d , it expands the exact identity

$$(4.2) \quad R_{u,v}(z^2) = z^d \tilde{R}_{u,v}(z - z^{-1})$$

over the integers. The result of (4.1) must correspond to the same unique normalized class as the result of (2.7), for every pair. This dual-recurrence check passes throughout.

The $n = 10$ run encounters 22 distinct normalized polynomials, with maximum coefficient 15. Its deterministic stream fingerprint is

0x78cf5914f6aaeb37.

The fingerprint is only a change detector, while SHA-256 hashes in the release manifest bind the actual source and input. The verifier also constructs, without using the observed class set as input, all product classes indexed by nondecreasing $h_i \geq 2$ with $\sum_i h_i \leq n - 2$ and requires exact equality with the observed class set in every rank $2 \leq n \leq 10$.

Proof of Theorem 1.1. The rank-matrix and subword constructions give the same complete universe of 12,866 elements. Testing all ordered pairs gives the complete set of 6,229,297 comparable pairs. Induction on the upper length via (2.7) reconstructs $B_{u,v}$ exactly for every pair. The finite generator bound $h \leq 17$ proves that the 434-element product catalog is exhaustive in the relevant degree. Every reconstructed polynomial belongs to the catalog. The independent ordinary recurrence and (4.2) agree with the same class at every induction node. Therefore (2.6), and hence the displayed formula in Theorem 1.1, holds for every quantified pair. $\square$

5. INDEPENDENT CHECKS AND REJECTION TESTS

The counterexample-search implementation is separate from the verifier. It generates the top interval from subwords, reconstructs it again by downward Bruhat-cover search, and enumerates each lower interval from a retained reduced word. Two full deterministic runs and a full Address-Sanitizer/UndefinedBehaviorSanitizer run agree exactly at $n = 10$. A Python implementation with full permutation enumeration, uncompressed q -polynomials, and exact polynomial division agrees through $n = 8$.

The standalone verifier regenerates all mathematical data and does not read the discovery outputs. Its closed-schema parser rejects missing, duplicate, or unknown fields; wrong and overflowing endpoints; a changed reduced word; a changed Fibonacci recurrence; and a false pair count. The destructive test suite first accepts the intact input and then requires every damaged input to terminate with nonzero status.

6. LIMITATIONS AND THE REMAINING STRUCTURAL PROBLEM

The 22 factor patterns seen at $n = 10$ are precisely those indexed by integer partitions with parts $h_i \geq 2$ and $\sum_i h_i \leq 8$. The analogous pattern holds in every computed rank. This is a certified finite observation at $n = 10$, not by itself an all-rank explanation.

A promising route uses distinguished masks. The exact recurrence identifies $B_{u,v}(x)$ with the defect-generating polynomial of distinguished masks ending at u . A general proof would follow if defect sets were the independent sets of a path forest and if different path components occupied disjoint footprints in the $n - 2$ column spine of Ω_n . The first statement would produce independent Fibonacci factors; the second would give $\sum h_i \leq n - 2$. Neither path-forest nor disjoint-footprint statement is proved here. They are the explicit gaps separating Theorem 1.1 from the all- n conjecture.

DATA, CODE, AND COMPUTATIONAL DISCLOSURE

The accompanying project contains the literal and formal statements, literature ledger, discovery programs and raw summaries, serialized certificate, standalone verifier, destructive-input tests, proof and referee audits, and a SHA-256 release manifest. The principal reproduction commands are

```
python3 certificates/verify_certificate.py
python3 tests/test_certificate_rejection.py
```

All decisive arithmetic is integral. No Lean, Coq, Isabelle, or other proof assistant was used. Generative-AI agents assisted with literature search, proof exploration, code construction, adversarial review, and exposition; every mathematical claim released here is limited to the exact statements and machine checks described above.

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